COMBINATORICS AND OPTIMIZATION

Notes

Fourth-year courses which require an 80% average as a prerequisite are held with corresponding graduate courses. Students with averages below 80% may enrol in these courses with the permission of the instructor.

CO 200s

CO 227 LEC 0.50

Course ID: 003887

Introduction to Optimization (Non-Specialist Level)

A broad introduction to the field of optimization, discussing applications, and solution techniques. Mathematical models for real life applications; algorithms: simplex, cutting plane, and branch & bound; linear programming duality. [Offered: F,W]

Prereq: One of MATH 106, 114, 115, 136, 146.

Antireq: CO 250, 255, 352

CO 250 LEC,TST,TUT 0.50

Course ID: 003895

Introduction to Optimization

A broad introduction to the field of optimization, discussing applications, and solution techniques. Mathematical models for real life applications; algorithms; aspects of computational complexity; geometry; linear programming duality, focusing on the development of algorithms. [Offered: F,W,S]

Prereq: One of (MATH 106, MATH 114, MATH 115 with a grade of at least 70%) or MATH 136 or MATH 146; cumulative overall average of at least 60%.

Antireq: CO 227, 255

Also offered Online

CO 255 LEC 0.50

Course ID: 003897

Introduction to Optimization (Advanced Level)

Linear optimization: feasibility theorems, duality, the simplex algorithm. Discrete optimization: integer linear programming, cutting planes, network flows. Continuous optimization: local and global optima, feasible directions, convexity, necessary optimality conditions.

[Note: CO 255 may be substituted for CO 250 whenever the latter is a requirement in an Honours plan. Offered: W]

Prereq: MATH 235 or 245, 237 or 247.

Antireq: CO 227, 250, 352

CO 300s

CO 327 LEC 0.50

Course ID: 003890

Deterministic OR Models (Non-Specialist Level)

An applications-oriented course that illustrates how various mathematical models and methods of optimization can be used to solve problems arising in business, industry, and science. [Offered: W,S]

Prereq: One of CO 227, 250, 255, 352.

Antireq: CO 370

CO 330 LEC 0.50

Course ID: 003891

Combinatorial Enumeration

The algebra of formal power series. The combinatorics of the ordinary and exponential generating series. Lagrange's implicit function theorem, applications to the enumeration of permutations, functions, trees and graphs. Integer partitions, geometric methods, enumerating linear transformations. Introduction to the pattern algebra, applications to the enumeration of strings. Lattice paths, Wiener-Hopf factorization. Enumeration under symmetries. [Offered: F]

Prereq: MATH 239 or 249

CO 331 LEC 0.50

Course ID: 003892

Coding Theory

A first course in error-correcting codes. Linear block codes, Hamming-Golay codes, and multiple error-correcting BCH codes are studied. Various encoding and decoding schemes are considered. [Offered: W]

Prereq: MATH 225 or 235 or 245

Only offered Online

CO 342 LEC 0.50

Course ID: 003893

Introduction to Graph Theory

An introduction to some of the key topics in graph theory: connectivity, planarity, and matchings. Connectivity: Menger's theorem, 3-connected graphs. Planarity: Kuratowski's theorem, uniqueness of planar embeddings. Matchings: Review of Konig's theorem, Tutte's theorem. [Offered: F,S]

Prereq: MATH 239 or 249

CO 351 LEC 0.50

Course ID: 003896

Network Flow Theory

Review of linear programming. Shortest path problems. The max-flow min-cut theorem and applications. Minimum cost flow problems. Network simplex and primal-dual algorithms. Applications to problems of transportation, distribution, job assignments, and critical-path planning. [Offered: F,S]

Prereq: One of CO 250 or 255 or 352

CO 353 LAB,LEC 0.50

Course ID: 011442

Computational Discrete Optimization

Formulations of combinatorial optimization problems, greedy algorithms, dynamic programming, branch-and-bound, cutting plane algorithms, decomposition techniques in integer programming, approximation algorithms.

Prereq: One of CO 250 or 255 or 352

CO 367 LAB,LEC 0.50

Course ID: 003898

Nonlinear Optimization

A course on the fundamentals of nonlinear optimization, including both the mathematical and the computational aspects. Necessary and sufficient optimality conditions for unconstrained and constrained problems. Convexity and its applications. Computational techniques and their analysis. [Offered: F]

Prereq: (One of CO 250, 255, 352) and MATH 128 with a grade of at least 70% or MATH 138 or 148

CO 370 LAB,LEC 0.50

Course ID: 003899

Deterministic OR Models

An applications-oriented course that illustrates how various mathematical models and methods of optimization can be used to solve problems arising in business, industry, and science. [Offered: F,W]

Prereq: CO 250 or 255 or 352.

Antireq: CO 327

CO 372 LAB,LEC 0.50

Course ID: 011736

Portfolio Optimization Models

Computational optimization methodologies underlying portfolio problems in finance. Computational linear algebra, determining derivatives, quadratic, and nonlinear optimization. The efficient frontier problem. Applications of optimization in finance such as volatility surface determination and global minimization for value-at-risk. [Offered: F,W]

Prereq: (AFM 272/ACTSC 291 or ACTSC 371 or BUS 393W or ECON 371) and (CO 227 with a grade of at least 70% or CO 250 or CO 255 or CO 352).

Antireq: CO 370 taken prior to winter 2004

CO 380 LEC 0.50

Course ID: 003901

Mathematical Discovery and Invention

A course in problem solving. 100 problems are studied. Problems are taken mainly from the elementary parts of algebra, geometry, number theory, combinatorics, and probability.

[Note: Offered in the spring term of even years.]

Prereq: MATH 135 or 145, MATH 106 or 136 or 146, MATH 138 or 148; Level at least 3A

CO 400s

CO 430 LEC 0.50

Course ID: 003902

Algebraic Enumeration

The Lagrange Implicit Function Theorem, the MacMahon Master Theorem. Enumeration of planar triangulations. The transfer matrix method. Sieve methods, inclusion/exclusion, Mobius inversion. Polya enumeration, Enumeration of trees. Basic hypergeometric series, q-analogues, Rogers-Ramanujan identities. Asymptotic methods.

Prereq: CO 330; Cumulative overall average of at least 80%

CO 431 LEC 0.50

Course ID: 016410

Symmetric Functions

The ring of symmetric functions, standard bases, the Hall inner product. Young tableaux. The Robinson-Schensted-Knuth correspondence, the hook-length formula, the Jacobi-Trudi formula, the Pieri rule, the Littlewood-Richardson rule. Representation theory of the symmetric groups. Enumeration of plane partitions. Enumeration of maps on surfaces. Other topics.

Prereq: PMATH 336 or PMATH 347; Cumulative overall average of at least 80%

CO 434 LEC 0.50

Course ID: 003903

Combinatorial Designs

Pairwise orthogonal latin squares. Transversal designs and finite planes. Balanced incomplete block designs, group divisible designs, and pairwise balanced designs. Symmetric designs and Hadamard matrices. Recursive constructions. Wilson's fundamental construction.

Prereq: PMATH 336 or 346 or 347; Cumulative overall average of at least 80%

CO 439 LEC 0.50

Course ID: 003906

Topics in Combinatorics

An undergraduate seminar in combinatorics. The primary objective is to study current work in specific areas of combinatorics. Course content may vary from term to term.

Instructor Consent Required

CO 440 LEC 0.50

Course ID: 003907

Topics in Graph Theory

An in-depth study of one or two topics in graph theory. Course content may vary from term to term. Topics may include planar graphs, extremal graph theory, directed graphs, enumeration, algebraic graph theory, probabilistic graph theory, connectivity, graph embedding, colouring problems.

Prereq: CO 342

CO 442 LEC 0.50

Course ID: 003908

Graph Theory

Connectivity (Menger's theorem, ear decomposition, and Tutte's wheels theorem) and matchings (Hall's theorem and Tutte's theorem). Flows: integer and group-valued flows, the flow polynomial, the 6-flow theorem. Ramsey theory: upper and lower bounds, explicit constructions. External graph theory: Turan's theorem, the Erdos-Gallai theorem. Probabilistic methods. [Offered: F]

Prereq: CO 342, MATH 235 or 245; Cumulative overall average of at least 80%

CO 444 LEC 0.50

Course ID: 003909

Algebraic Graph Theory

An introduction to the methods of and some interesting current topics in algebraic graph theory. Topics covered will include vertex-transitive graphs, eigenvalue methods, strongly regular graphs and may include graph homomorphisms, Laplacians or knot and link invariants.

Prereq: MATH 239 or 249, PMATH 336 or 346 or 347; Cumulative overall average of at least 80%

CO 446 LEC 0.50

Course ID: 013337

Matroid Theory

This is an introductory course on matroid theory, with particular emphasis on graphic matroids and on topics that are applicable to graph theory. The topics include matroid intersection and partition, graphic matroids, matroid connectivity, regular matroids, and representable matroids. Applications include matching, disjoint paths, disjoint spanning trees, the 8-flow theorem for graphs, planarity testing, and recognizing totally unimodular matrices. [Offered: S]

Prereq: CO 342; Cumulative overall average of at least 80%

CO 450 LEC 0.50

Course ID: 003910

Combinatorial Optimization

Characterizations of optimal solutions and efficient algorithms for optimization problems over discrete structures. Topics include network flows, optimal matchings, T-joins and postman tours, matroid optimization. [Offered: F]

Prereq: CO 255 or 351; Cumulative overall average of at least 80%

CO 452 LEC 0.50

Course ID: 003911

Integer Programming

Formulation of problems as integer linear programs. Solution by branch-and-bound and cutting plane algorithms. Introduction to the theory of valid inequalities and polyhedral combinatorics.

Prereq: CO 255 or 351; Cumulative overall average of at least 80%

CO 454 LEC 0.50

Course ID: 003913

Scheduling

An overview of practical optimization problems that can be posed as scheduling problems. Characterizations of optimal schedules. Simple and efficient combinatorial algorithms for easy problems. A brief overview of computational complexity, definition of P, NP, NP-complete and NP-hard. Integer programming formulations, the traveling salesman problem, heuristics, dynamic programming, and branch-and-bound approaches. Polynomial-time approximation algorithms. [Offered: S]

Prereq: MATH 229 or 239 or 249 and (one of CO 227, 250, 255, 352)

CO 456 LEC 0.50

Course ID: 003914

Introduction to Game Theory

A broad introduction to game theory and its applications to the modeling of competition and cooperation in business, economics, and society. Two-person games in strategic form and Nash equilibria. Extensive form games, including multi-stage games. Coalition games and the core. Bayesian games, mechanism design, and auctions.

Prereq: MATH 229 or 239 or 249 and (one of CO 227, 250, 255, 352)

CO 459 SEM 0.50

Course ID: 010046

Topics in Optimization

An undergraduate seminar in optimization. The primary objective is to study recent work in specific areas of optimization. Course content may vary from term to term.

Instructor Consent Required

CO 463 LEC 0.50

Course ID: 010047

Convex Optimization and Analysis

An introduction to the modern theory of convex programming, its extensions and applications. Structure of convex sets, separation and support, subgradient calculus for convex functions, Fenchel conjugacy and duality, Lagrange multipliers. Ellipsoid method for convex optimization. [Offered: W]

Prereq: (CO 255 or 367), (AMATH/PMATH 331 or PMATH 351); Cumulative overall average of at least 80%

CO 466 LEC 0.50

Course ID: 003917

Continuous Optimization

Numerical algorithms for nonlinear optimization. Newton, variable-metric, quasi-Newton and conjugate gradient methods. Obtaining derivatives. Convexity. Trust region methods. Constrained optimization including optimality conditions, sequential quadratic programming, interior point, and active set strategies.

Prereq: (CO 367 and one of CO 250, 352) or CO 255; Cumulative overall average of at least 80%

CO 471 LEC 0.50

Course ID: 011364

Semidefinite Optimization

Optimization over convex sets described as the intersection of the set of symmetric, positive semidefinite matrices with affine spaces. Formulations of problems from combinatorial optimization, graph theory, number theory, probability and statistics, engineering design, and control theory. Theoretical and practical consequences of these formulations. Duality theory and algorithms.

Prereq: MATH 239 or 249, AMATH/PMATH 331 or PMATH 351, CO 255; Cumulative overall average of at least 80%

CO 480 LEC 0.50

Course ID: 003918

History of Mathematics

An in-depth examination of the origins of mathematics, beginning with examples of Babylonian mathematics. Topics may include Pythagorean triples, solution of equations, estimation of pi, duplication of the cube, trisection of an angle, the Fibonacci sequence, the origins of calculus.

[Note: Offered in the spring term of odd years.]

Prereq: MATH 135 or 145, MATH 106 or 136 or 146, MATH 138 or 148; Level at least 3A

CO 481 LEC,TST 0.50

Course ID: 011497

Introduction to Quantum Information Processing

Basics of computational complexity; basics of quantum information; quantum phenomena; quantum circuits and universality; relationship between quantum and classical complexity classes; simple quantum algorithms; quantum Fourier transform; Shor factoring algorithm; Grover search algorithm; physical realization of quantum computation; error-correction and fault-tolerance; quantum key distribution. [Offered: F, W]

Prereq: One of MATH 114, 115, 235, 245; Level at least 4A

(Cross-listed with CS 467, PHYS 467)

CO 485 LEC 0.50

Course ID: 010137

The Mathematics of Public-Key Cryptography

An in-depth study of public-key cryptography. Number-theoretic problems: prime generation, integer factorization, discrete logarithms. Public-key encryption, digital signatures, key establishment, elliptic curve cryptography, post-quantum cryptography. Proofs of security. [Offered: F]

Prereq: One of PMATH 334, 336, 345, 346, 347; Cumulative overall average of at least 80%

CO 487 LEC 0.50

Course ID: 010136

Applied Cryptography

A broad introduction to modern cryptography, highlighting the tools and techniques used to secure internet and messaging applications. Symmetric-key encryption, hash functions, message authentication, authenticated encryption, public-key encryption and digital signatures, key establishment, key management. [Offered: F,W]

Prereq: One of CS 116, 136, 138, 146; MATH 135 or 145; One of STAT 206, 220, 230, 240; Level at least 3A

CO 499 RDG 0.50

Course ID: 003920

Reading in Combinatorics and Optimization

Reading course as announced by the department.

Department Consent Required